3ART13 LED

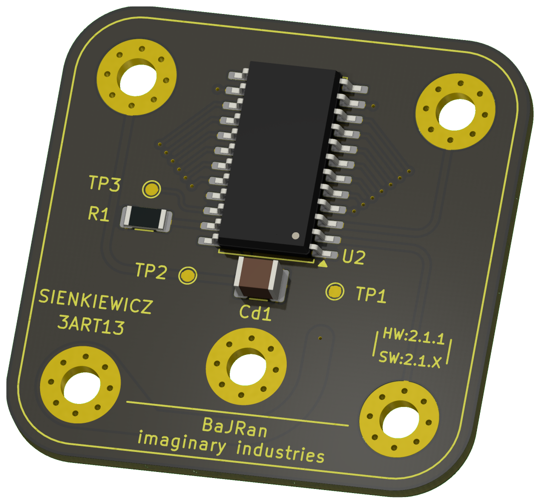
Variant: RELEASED

2026-05-21

Rev 2.1.1

TOP VIEW

BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 21-May-2026

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

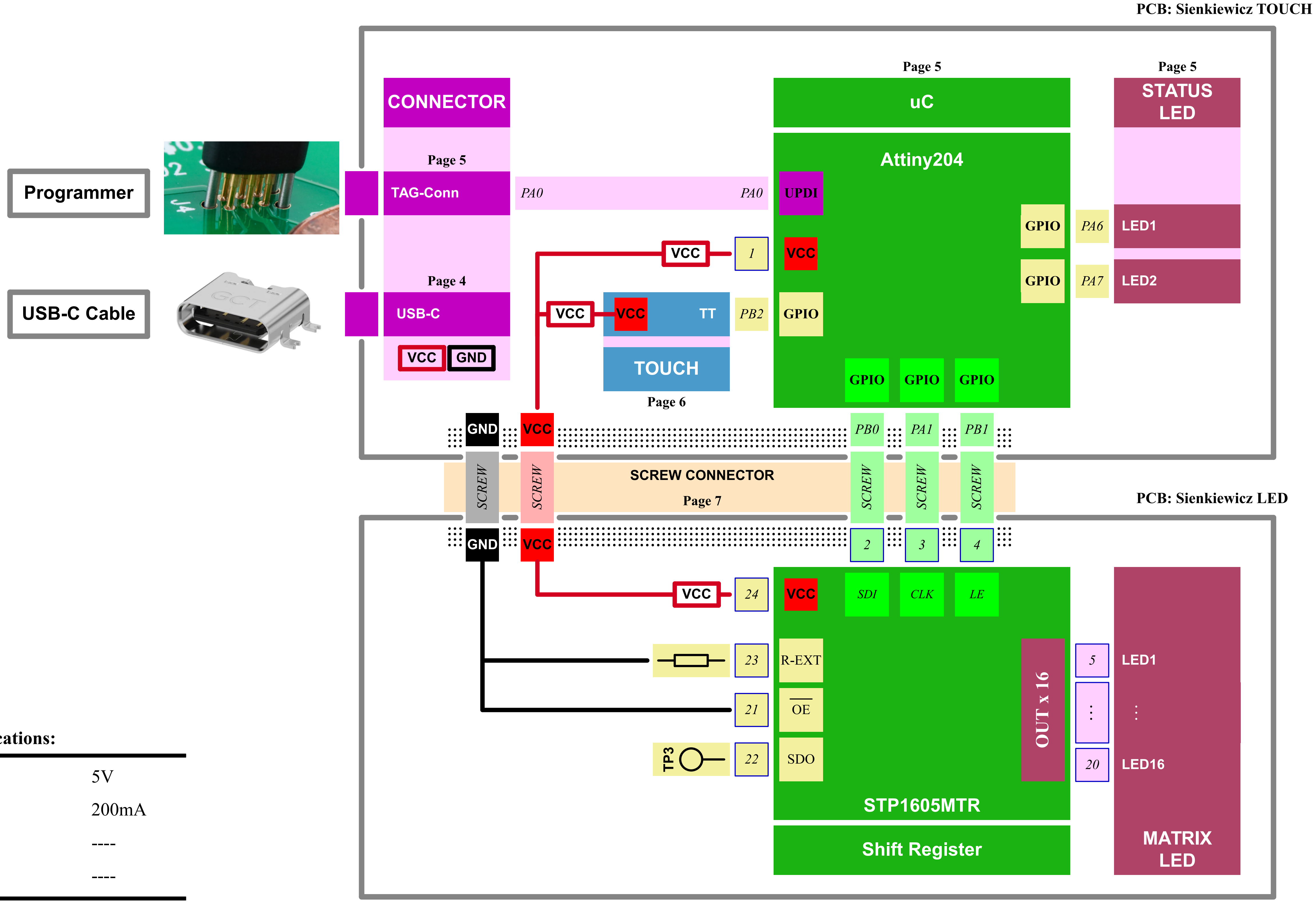
DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

Comments:	Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: e1012a8
	Board Name: 3ART13 LED		Project Name: Sienkiewicz	
Sheet Title:	File Name: Sienkiewicz_LED.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.1
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 8

[2] Block Diagram

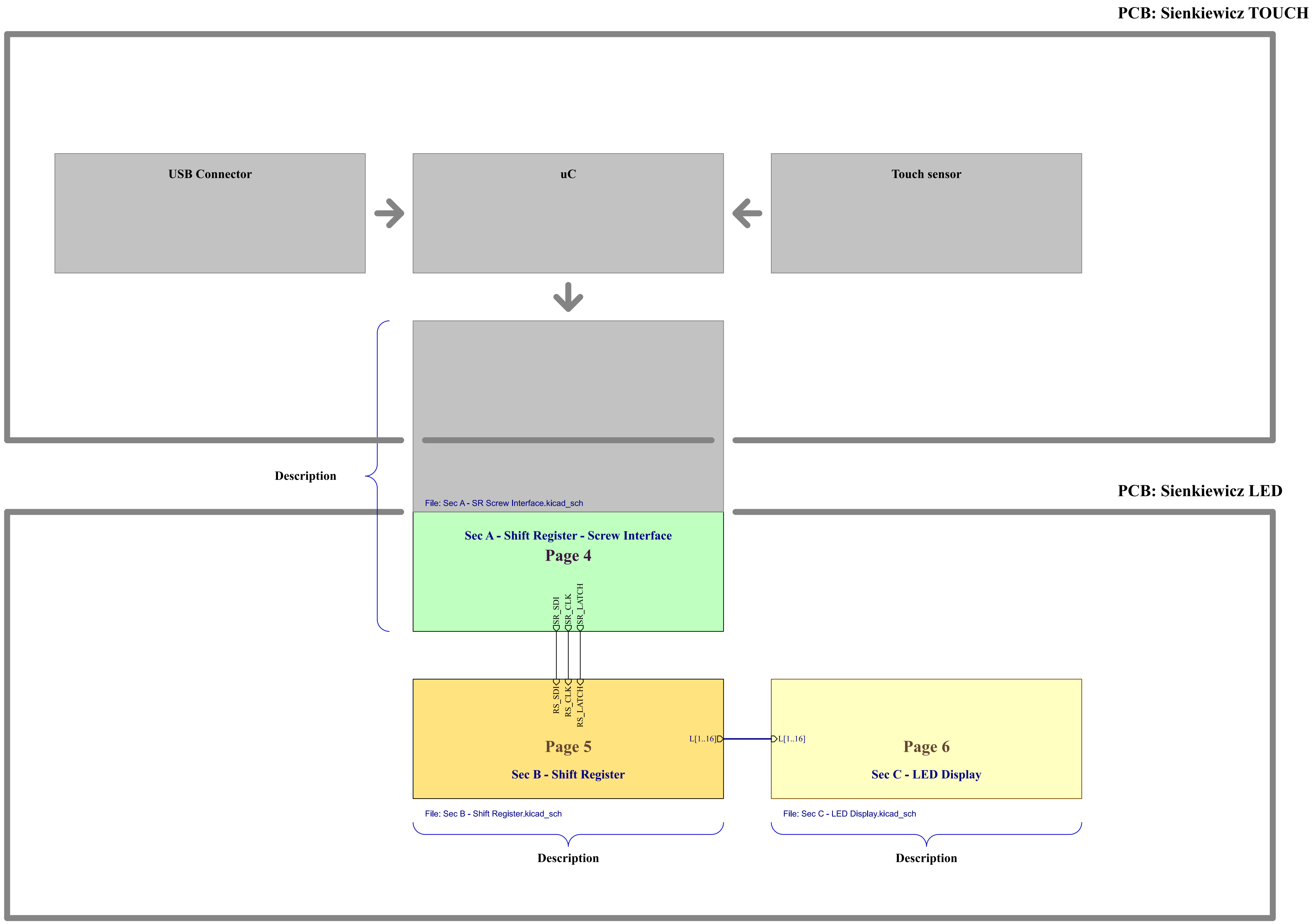


Target specifications:

Input voltage	5V
Current	200mA
Spec 3	----
Spec 4	----

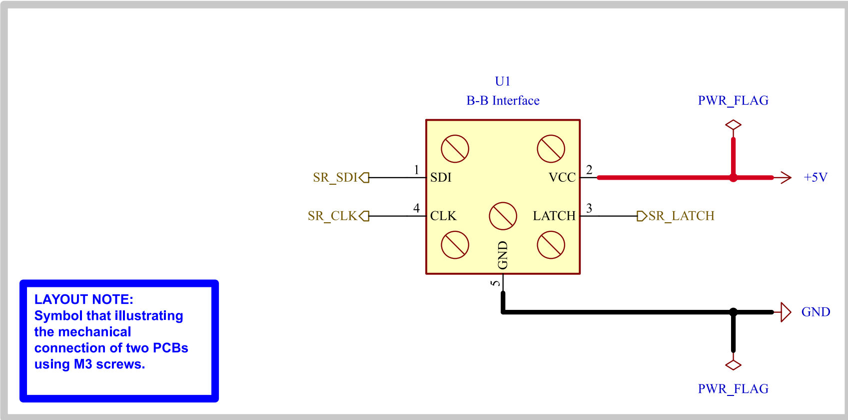
Comments:	Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: e1012a8
	Board Name: **3ART13 LED**		Project Name: **Sienkiewicz**	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.1
Sheet Path: /Block Diagram/		Reviewer:	Size: **A3**	Sheet: **2** of **8**

[3] Project Architecture



Comments:	Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: e1012a8	
	Board Name: 3ART13 LED		Project Name: Sienkiewicz		
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.1	
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 8	

[4] Shift register - Screw Interface



Comments:
AN5346
STM32G474 Datasheet p.81
J. Pieper ADC investigation

Company: **BaJRan imgaginary industries**

Variant: RELEASED

Git Hash: e1012a8

Board Name: **3ART13 LED**

Project Name: **Sienkiewicz**

Sheet Title: Shift register - Screw Interface

File Name: Sec A - SR Screw Interface.kicad_sch

Designer: Pawel Baran

Date: 2025-01-12

Revision: 2.1.1

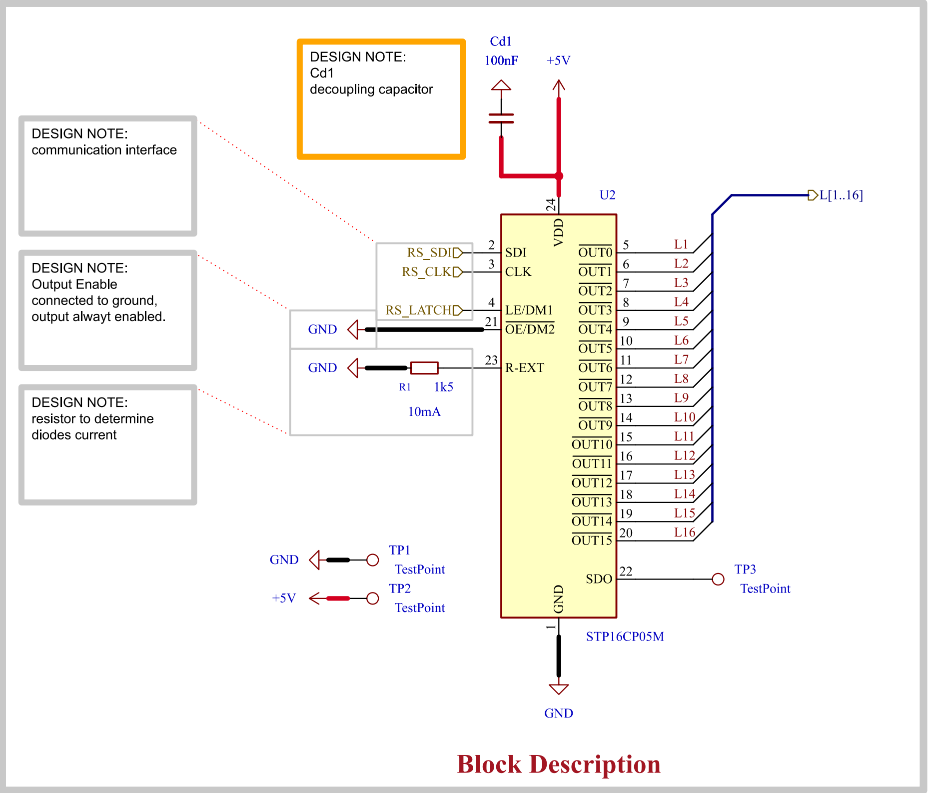
Sheet Path: /Project Architecture/Sec A - Shift Register - Screw Interface/

Reviewer:

Size: **A4**

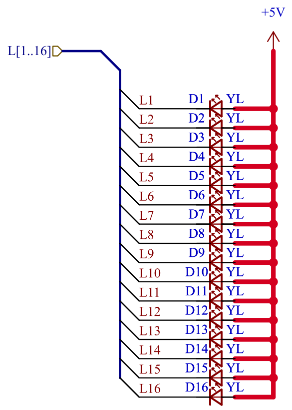
Sheet: **4** of **8**

[5] Shift Register



Comments:	Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: e1012a8
	Board Name: 3ART13 LED		Project Name: Sienkiewicz	
Sheet Title: Shift Register	File Name: Sec B - Shift Register.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.1
Sheet Path: /Project Architecture/Sec B - Shift Register/		Reviewer:	Size: A4	Sheet: 5 of 8

[6] LED Display



Block Description

Comments:

Company: **BaJRan imgaginary industries**

Variant: RELEASED

Git Hash: e1012a8

Board Name: **3ART13 LED**

Project Name: **Sienkiewicz**

Sheet Title: LED Display

File Name: Sec C - LED Display.kicad_sch

Designer: Pawel Baran

Date: 2026-04-30

Revision: 2.1.1

Sheet Path: /Project Architecture/Sec C - LED Display/

Reviewer:

Size: **A4**

Sheet: **6** of **8**

[7] Requirements

Project	Simplified electronic project designed to verify practical skills in electrical engineering, PCB design, electronic assembly, embedded software implementation, and programming within the Microchip Technology development environment. Simple powering.	MEET	The project is a simple LED light effect device with a user interface realized using a capacitive touch sensor. Powered by USB-C Connector.
MAIN OBJECTIVE			

1	The project shall contain very small integrated circuits. Pitch less than 1mm	MEET	IC: Cap sensor : AT42QT1012 : SOT-23-6 package : PITCH: 0.95 IC: uC : Attiny-204 : SOIC-14_3.9x8.7mm_P1.27mm : PITCH 1.27
2	The project shall contain 0805 SMT passives components.	MEET	Device contain multiple 0805 SMT elements.
3	The project shall contain multi-pin integrated circuits to verify soldering precision.	MEET	IC: Shift register : STP16CP05MTR : SO-24 : 24 pins IC: uC : Attiny204 : SOIC-14_3.9x8.7mm : 14 pins
4	The project shall contain an UPDI programming interface for firmware upload and debugging.	MEET	uC Attiny204 with UPDI programming interface
5	Possibility to program using Microchip enviroment and pickit-4 programmer.	MEET	It is possible to program the device using pickit-4 and MPLAB software.
6	Shall contain at least two status led.	MEET	Status LED'S : LED1, LED2.
7	Shall be able to interact with user.	MEET	The user can change the light effect using a capacitive touch sensor.
8	Shall be easy to power up.	MEET	Powered by USB-C connector with general mode
9			
10			
11			
12			

Comments:	Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: e1012a8
	Board Name: 3ART13 LED		Project Name: Sienkiewicz	
Sheet Title: Requirements	File Name: Power - Sequencing.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.1
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 7 of 8

[8] Revision History

Version 2.1.0 - 2026-05-02

Added

- First version of PCB and Schematic

Version 2.1.1 - 2026-05-21

Comments:	Company: BaJRan imgaginary industries		Variant: RELEASED	Git Hash: e1012a8
	Board Name: 3ART13 LED		Project Name: Sienkiewicz	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Pawel Baran	Date: 2025-01-12	Revision: 2.1.1
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 8 of 8